

Conic Sections: Parabolas

Analytic Geometry Worksheet · Grade 10–12

Name: _____

Date: _____

Learning Objectives

- Identify the vertex, focus, and directrix of a parabola from its equation or graph
- Write the standard equation of a parabola opening vertically or horizontally
- Determine the direction a parabola opens based on the sign of p

Problems

1. A parabola has its vertex at the origin and opens upward with $p = 2$. Write its standard equation.

$$(x - 0)^2 = 4(2)(y - 0)$$

2. Identify the direction a parabola opens if its equation is given below and p is negative.

$$(x - h)^2 = 4p(y - k), \quad p < 0$$

3. A parabola opens to the right with vertex at $(0, 0)$ and $p = 3$. Write its standard equation.

$$(y - k)^2 = 4p(x - h)$$

4. Find the focus and directrix of the parabola given below.

$$x^2 = 16y$$

5. Find the vertex, focus, and directrix of the parabola given below.

$$(x - 2)^2 = -8(y - 3)$$

6. Find the vertex, focus, and directrix of the parabola given below.

$$(y + 1)^2 = -12(x - 4)$$



7. Write the standard equation of a parabola with vertex at (3, -2) and focus at (3, 2).

$$(x - h)^2 = 4p(y - k)$$

8. Convert the equation below to standard form and identify the vertex, focus, and directrix.

$$x^2 - 6x - 8y + 1 = 0$$

9. A parabola has its vertex at (-1, 4) and its directrix at $x = 2$. Write its standard equation.

$$(y - k)^2 = 4p(x - h)$$

10. Convert the equation below to standard form, then find the vertex, focus, directrix, and axis of symmetry.

$$y^2 + 10y - 4x + 29 = 0$$



Conic Sections: Parabolas — Answer Key

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Answer Key

1. Answer: $x^2 = 8y$

- The parabola opens vertically upward, so use $(x - h)^2 = 4p(y - k)$.
- Vertex is $(0, 0)$, so $h = 0$ and $k = 0$.
- $p = 2$, so $4p = 8$.
- Standard equation: $x^2 = 8y$.

2. Answer: Opens downward

- The equation $(x - h)^2 = 4p(y - k)$ represents a vertical parabola.
- When $p > 0$, it opens upward; when $p < 0$, it opens downward.
- Since p is negative, the parabola opens downward.

3. Answer: $y^2 = 12x$

- The parabola opens horizontally to the right, so use $(y - k)^2 = 4p(x - h)$.
- Vertex is $(0, 0)$, so $h = 0$ and $k = 0$.
- $p = 3$, so $4p = 12$.
- Standard equation: $y^2 = 12x$.

4. Answer: Focus: $(0, 4)$; Directrix: $y = -4$

- Compare $x^2 = 16y$ with $x^2 = 4py$: $4p = 16$, so $p = 4$.
- Since $p > 0$ the parabola opens upward with vertex at $(0, 0)$.
- Focus is at $(h, k + p) = (0, 0 + 4) = (0, 4)$.
- Directrix is $y = k - p = 0 - 4 = -4$.

5. Answer: Vertex: $(2, 3)$; Focus: $(2, 1)$; Directrix: $y = 5$

- Vertex is $(h, k) = (2, 3)$.
- $4p = -8$, so $p = -2$. Negative p means it opens downward.
- Focus is at $(h, k + p) = (2, 3 + (-2)) = (2, 1)$.
- Directrix is $y = k - p = 3 - (-2) = 5$.

6. Answer: Vertex: $(4, -1)$; Focus: $(1, -1)$; Directrix: $x = 7$

- Vertex is $(h, k) = (4, -1)$.
- $4p = -12$, so $p = -3$. Negative p means it opens to the left.
- Focus is at $(h + p, k) = (4 + (-3), -1) = (1, -1)$.
- Directrix is $x = h - p = 4 - (-3) = 7$.

7. Answer: $(x - 3)^2 = 16(y + 2)$

- Vertex is $(3, -2)$ and focus is $(3, 2)$; same x -coordinate means vertical parabola.
- $p =$ distance from vertex to focus $= 2 - (-2) = 4$.
- $4p = 16$



- Equation: $(x - 3)^2 = 16(y + 2)$.

8. Answer: $(x - 3)^2 = 8(y + 1)$; Vertex: (3, -1); Focus: (3, 1); Directrix: $y = -3$

- Group and complete the square: $(x^2 - 6x) = 8y - 1$.
- Complete the square: $(x - 3)^2 - 9 = 8y - 1$.
- $(x - 3)^2 = 8y - 1 + 9 = 8y + 8 = 8(y + 1)$.
- Standard form: $(x - 3)^2 = 8(y + 1)$; vertex (3, -1), $p = 2$, focus (3, 1), directrix $y = -3$.

9. Answer: $(y - 4)^2 = -12(x + 1)$

- The directrix is a vertical line $x = 2$, so the parabola opens horizontally.
- Vertex is (-1, 4). The directrix $x = 2$ is to the right of vertex, so it opens left.
- $p =$ distance from vertex to directrix $= -1 - 2 = -3$.
- $4p = -12$. Equation: $(y - 4)^2 = -12(x + 1)$.

10. Answer: $(y + 5)^2 = 4(x - 1)$; Vertex: (1, -5); Focus: (2, -5); Directrix: $x = 0$; Axis: $y = -5$

- Group y terms: $(y^2 + 10y) = 4x - 29$.
- Complete the square: $(y + 5)^2 - 25 = 4x - 29$.
- $(y + 5)^2 = 4x - 29 + 25 = 4x - 4 = 4(x - 1)$.
- Standard form: $(y + 5)^2 = 4(x - 1)$; vertex (1, -5), $4p = 4$ so $p = 1$.
- Focus: $(1 + 1, -5) = (2, -5)$. Directrix: $x = 1 - 1 = 0$. Axis of symmetry: $y = -5$.

